

1. Introduction

Overview

Centra-Experitest is an automated application monitoring and Quality of Experience (QoE) assessment platform designed for continuous monitoring of mobile and web-based applications. The solution integrates Centra-SD with Digital.ai SeeTest Cloud to automate application testing, monitor network performance, detect service anomalies, and provide proactive alerts before end users experience service degradation.

Traditional network monitoring solutions primarily evaluate network-level parameters such as signal strength, throughput, latency, and protocol performance. However, modern mobile services depend equally on the performance of applications running over these networks. Even when network KPIs appear satisfactory, users may experience poor service due to application failures, slow loading times, or backend issues.

Centra-Experitest addresses this challenge by continuously executing automated test scenarios on real smartphones and web platforms. These automated tests simulate actual user interactions with applications and websites, providing organizations with accurate insight into the end-user experience.

The platform operates continuously throughout the year, enabling 24×7×365 monitoring of critical services. Whenever predefined performance thresholds are exceeded or application failures are detected, Centra-Experitest generates alerts, allowing operators to investigate and resolve issues before customers begin reporting complaints.

By combining automated testing, centralized monitoring, and intelligent reporting, Centra-Experitest enables telecom operators, enterprises, and application developers to improve service quality while significantly reducing operational costs.

2. Purpose of Centra-Experitest

The primary objective of Centra-Experitest is to automate application monitoring and provide continuous visibility into the quality of digital services experienced by end users.

Instead of relying on manual testing or customer feedback, the platform executes predefined application workflows at scheduled intervals and evaluates the results automatically.

The main objectives include:

- Continuous monitoring of mobile applications
- Continuous monitoring of web applications
- Early detection of application failures

- Automatic execution of application test scenarios
- Measurement of Quality of Experience (QoE)
- Network and application performance visualization
- Reduction of manual testing effort
- Improvement of customer satisfaction
- Support for proactive network maintenance
- Collection of long-term application performance data

By performing automated testing on real devices, the platform provides realistic measurements that closely reflect the actual user experience.

3. Product Objectives

Centra-Experitest has been developed to achieve several strategic goals within network operations and digital service management.

Improve Customer Experience

Monitor applications continuously to identify performance degradation before customers encounter issues.

Reduce Operational Costs

Replace repetitive manual testing with automated monitoring workflows, reducing engineering effort and operational expenses.

Detect Problems Early

Automatically identify service anomalies and notify engineers immediately, enabling proactive issue resolution.

Support Multiple Platforms

Monitor both native mobile applications and browser-based web applications using a unified testing platform.

Increase Monitoring Coverage

Execute automated tests throughout the day and night, ensuring continuous visibility into service quality.

Enable Scalable Testing

Support execution of automated scripts across large fleets of real devices to accommodate enterprise-scale deployments.

4. System Architecture

Centra-Experitest combines automated application testing with centralized monitoring and reporting. The architecture allows centralized scheduling and monitoring while leveraging real devices for realistic application testing.

5. Working Principle

Centra-Experitest operates by executing automated test scripts on real devices according to predefined schedules. The results are analyzed and visualized through Centra-SD.

The workflow consists of the following stages:

Step 1 – Test Scenario Creation

Engineers create automated workflows that simulate user interactions with an application or website.

Examples include:

- Login
- Search
- Product Selection
- Video Playback
- File Upload
- Payment Process
- Logout

These workflows represent real customer behavior and provide meaningful performance measurements.

Step 2 – Schedule Configuration

The test scenarios are assigned to schedules within Centra-SD.

Schedules define:

- Execution frequency
 - Test duration
 - Device assignment
 - Target application
 - Network conditions
 - Alert thresholds
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Step 3 – Automated Test Execution

At the scheduled time, Digital.ai SeeTest Cloud executes the test scripts on real smartphones or web browsers without requiring manual intervention.

During execution, the system records:

- Application launch time
 - Response time
 - Screen transitions
 - Network performance
 - Application crashes
 - Script execution status
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Step 4 – Performance Evaluation

After each test, Centra-Experitest compares the collected metrics with predefined performance thresholds.

Examples include:

- Login completed within 5 seconds
- Video started within 3 seconds
- Payment completed successfully
- Page loaded within acceptable response time

If the observed values exceed the configured limits, the system classifies the event as a potential service anomaly.

Step 5 – Alert Generation

Whenever a service anomaly is detected, Centra-Experitest automatically generates alerts and forwards them to Centra-SD.

Notifications may include:

- Application failure
- Login failure
- Slow response
- Video buffering
- Network timeout
- Unexpected application crash
- Script execution failure

These alerts enable network operations teams to investigate problems before they impact a significant number of users.

6. Key Characteristics

Centra-Experitest provides several distinguishing characteristics that make it suitable for enterprise-scale application monitoring.

Automated Testing

All monitoring activities are performed automatically according to predefined schedules, eliminating repetitive manual testing.

Real Device Testing

Unlike simulator-based solutions, tests are executed on real smartphones, providing accurate measurements that closely represent actual user experiences.

Continuous Monitoring

The platform operates continuously throughout the year, allowing organizations to monitor critical applications without interruption.

Integration with Centra-SD

All measurement results are centralized within Centra-SD, enabling unified visualization, reporting, and historical analysis.

Scalable Architecture

The platform supports execution of automated scripts across thousands of connected devices, making it suitable for large telecom operators and enterprise deployments.

Cross-Platform Support

Centra-Experitest supports both mobile applications and browser-based web applications, allowing organizations to monitor multiple digital services using a single platform.

7. Automation Engine

Automation is the foundation of Centra-Experitest. Modern mobile applications and web services are expected to be available at all times, making manual monitoring impractical. Centra-Experitest addresses this challenge by automatically executing predefined test scenarios on real devices according to configurable schedules.

Instead of requiring engineers to manually launch applications and verify functionality, the platform performs these tasks automatically, records the results, and reports any deviations from expected behavior.

Automation enables organizations to monitor services continuously, ensuring that performance issues are identified as soon as they occur.

Benefits of Automation

- Eliminates repetitive manual testing
- Provides consistent and repeatable test execution
- Supports 24×7×365 monitoring
- Reduces operational costs
- Increases monitoring frequency
- Enables proactive maintenance
- Improves service reliability

Automation also ensures that every execution follows the same sequence of actions, making performance comparisons more reliable over time.

7.1 Scheduled Test Execution

Centra-Experitest allows administrators to define test schedules based on operational requirements.

Schedules may include:

- Every 5 minutes
- Every 30 minutes
- Hourly
- Daily
- Weekly
- Continuous execution

Each schedule can be associated with:

- Specific mobile applications
- Web applications
- Device groups
- Geographic locations
- Network technologies
- Business hours

- Peak traffic periods

Once configured, schedules are automatically distributed to the assigned devices.

7.2 Background Monitoring

One of the key advantages of Centra-Experitest is its ability to perform monitoring in the background without requiring constant human supervision.

Background monitoring allows organizations to verify application availability even during non-business hours.

Examples include:

- Banking applications
- Video streaming platforms
- Customer self-service portals
- Enterprise applications
- Government service portals
- Healthcare applications

This ensures uninterrupted visibility into service availability.

8. Digital.ai SeeTest Cloud Integration

Centra-Experitest integrates with **Digital.ai SeeTest Cloud**, a widely used mobile application testing platform that enables automated testing on real smartphones and tablets.

Through this integration, organizations can execute automated scripts on multiple devices while Centra-SD manages scheduling, monitoring, and reporting.

8.1 Real Device Testing

Unlike emulator-based testing, SeeTest Cloud executes scripts on physical smartphones.

Benefits include:

- Accurate hardware performance
- Real network behavior
- Genuine application rendering
- Reliable touch interactions
- Camera and sensor compatibility
- Real operating system behavior

Testing on real devices provides a more realistic representation of the end-user experience.

8.2 Multi-Platform Support

The integrated solution supports testing across a variety of platforms, including:

- Android
- iOS
- Web browsers
- Progressive Web Applications (PWAs)
- Hybrid applications

This flexibility enables organizations to validate application performance across different operating systems and device types.

8.3 Large-Scale Device Support

Digital.ai SeeTest Cloud supports execution of automated scripts across **1000+ devices**.

This capability enables:

- Parallel testing
- Device comparison
- Cross-platform validation
- Large-scale regression testing
- Nationwide monitoring

Parallel execution significantly reduces overall testing time while increasing coverage.

9. Quality of Experience (QoE)

Traditional monitoring focuses primarily on network-level indicators such as signal strength and throughput. However, users evaluate services based on their overall experience when interacting with applications.

Centra-Experitest measures **Quality of Experience (QoE)** by evaluating how applications perform from the perspective of a real user.

QoE Parameters

Examples of QoE measurements include:

- Application launch time

- Login success
- Page loading time
- Screen transition time
- Video startup delay
- Buffering duration
- Download completion
- Payment transaction success
- Form submission response
- Application crash frequency

These measurements provide direct insight into the usability of digital services.

Why QoE Matters

Even if network KPIs indicate acceptable performance, poor application responsiveness can still result in customer dissatisfaction.

QoE monitoring helps organizations:

- Detect slow applications
 - Improve customer satisfaction
 - Reduce service disruptions
 - Increase user retention
 - Validate software releases
 - Verify infrastructure upgrades
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10. Alarm and Notification System

Centra-Experitest continuously evaluates collected measurements against predefined thresholds.

Whenever abnormal behavior is detected, the system generates alerts that allow engineers to respond immediately.

Alarm Types

Examples include:

- Application unavailable
- Login failed
- Server timeout

- Slow response time
 - Network connectivity issue
 - Application crash
 - Script execution failure
 - Authentication error
 - Video playback interruption
 - Download failure
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Alert Workflow

The alarm management process typically follows these steps:

1. Automated test execution.
2. Measurement collection.
3. KPI evaluation.
4. Threshold comparison.
5. Alarm generation.
6. Notification delivery.
7. Investigation by engineering teams.
8. Issue resolution.
9. Verification through subsequent automated tests.

This proactive workflow helps organizations resolve issues before customers begin reporting problems.

11. Performance Visualization

Centra-Experitest provides visualization tools that transform measurement data into meaningful insights.

Visualization enables engineers to quickly understand service health without manually examining large log files.

Dashboard View

The dashboard provides an overview of:

- Running tests
- Successful tests

- Failed tests
- Average response time
- Current application status
- Active alarms
- Device availability

The dashboard is updated continuously as new measurements are received.

Charts and Graphs

Performance metrics can be displayed using:

- Line Charts
- Bar Charts
- Pie Charts
- Trend Graphs
- Comparative Graphs

Graphical visualization simplifies long-term performance analysis and capacity planning.

Speed Test Visualization

Centra-Experitest can also monitor publicly available speed test applications to evaluate network performance.

Typical measurements include:

- Download Speed
- Upload Speed
- Latency
- Network Stability
- Response Time

These results help correlate application performance with underlying network conditions.

12. Script Management

Automated testing is driven by scripts that simulate user behavior.

Centra-Experitest allows engineers to create, manage, and execute custom scripts tailored to different applications.

Custom Test Scenarios

Organizations can create workflows representing real customer interactions.

Examples include:

Banking Application

- Launch application
 - Enter credentials
 - Login
 - Check account balance
 - Transfer funds
 - Logout
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E-Commerce Application

- Open application
 - Search for product
 - View product details
 - Add to cart
 - Proceed to checkout
 - Complete payment
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Video Streaming Platform

- Launch application
 - Search for video
 - Start playback
 - Pause playback
 - Resume playback
 - Stop playback
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Government Portal

- Login
- Search services
- Submit application

- Download receipt

These realistic workflows allow organizations to evaluate complete business processes rather than isolated technical parameters.

Script Reusability

Once created, scripts can be reused across:

- Different devices
- Multiple operating systems
- Various testing campaigns
- Regression testing
- Acceptance testing

Reusable scripts reduce development effort and ensure standardized testing procedures.

13. KPI Monitoring

Centra-Experitest records a wide range of application performance indicators that help organizations evaluate service quality.

Examples include:

Application KPIs

- Launch Time
 - Login Time
 - Response Time
 - Crash Rate
 - Success Rate
 - Error Count
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Network KPIs

- Download Speed
 - Upload Speed
 - Latency
 - Packet Loss
 - Signal Quality
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Device KPIs

- Battery Level
 - CPU Usage
 - Memory Usage
 - Storage Availability
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Business KPIs

- Transaction Success Rate
- Average User Journey Time
- Failed Transactions
- Service Availability
- SLA Compliance

By combining technical and business metrics, Centra-Experitest provides a comprehensive view of application performance and user experience.

14. Synchronized Monitoring

A notable feature of Centra-Experitest is synchronized monitoring, where multiple devices execute the same test scenario simultaneously.

This enables engineers to compare performance across:

- Different operators
- Different cities
- Different device models
- Different operating systems
- Different network technologies

Synchronized execution helps identify whether performance issues are localized or widespread, improving the efficiency of troubleshooting and benchmarking activities.

15. Reporting and Log Analysis

Centra-Experitest generates detailed reports after every automated test execution. These reports provide valuable insights into application behavior, network performance, and user experience. By analyzing these reports, engineers can identify performance bottlenecks, verify service availability, and ensure that applications meet predefined quality standards.

Each report contains detailed execution logs, timestamps, device information, network conditions, and application-specific metrics. This information enables engineering teams to investigate failures efficiently and compare performance across different testing periods.

Information Included in Reports

A typical report may contain:

- Test execution status
- Application launch time
- Login response time
- Page loading time
- Network latency
- Download and upload speeds
- Device information
- Operating system version
- Test execution duration
- Error messages
- Screenshot references (if configured)
- Alarm status
- Script execution logs

Reports can be viewed through the Centra-SD interface or exported for offline analysis and documentation.

16. Detailed Log Analysis

One of the most powerful capabilities of Centra-Experitest is its ability to generate detailed execution logs. These logs record every action performed during automated testing and provide engineers with complete visibility into the application's behavior.

Each interaction performed by the automated script is timestamped and recorded, allowing engineers to reconstruct the entire user journey.

Examples of Logged Events

- Application launched
- Login button clicked
- Username entered
- Password submitted
- Authentication successful

- Dashboard loaded
- Search initiated
- Video playback started
- Network timeout detected
- Application terminated
- Test completed successfully

When failures occur, the log provides sufficient information to identify the exact step where the problem occurred.

Error Identification

Detailed logs help identify various categories of application failures, including:

Functional Errors

- Login failure
- Transaction failure
- Incorrect application response
- Missing user interface elements

Performance Errors

- Slow application launch
- Delayed screen transitions
- Long server response time
- Excessive buffering

Network Errors

- Connection timeout
- DNS resolution failure
- Packet loss
- Poor signal quality

Device Errors

- Low battery
- Storage limitations
- Application crash
- Operating system incompatibility

By classifying failures into these categories, engineers can determine whether the root cause lies within the application, network, backend infrastructure, or device environment.

17. Historical Performance Analysis

Centra-Experitest maintains historical records of all executed tests, enabling organizations to analyze application performance over extended periods.

Historical analysis provides valuable information regarding long-term trends and recurring service issues.

Benefits of Historical Analysis

- Compare current and previous performance
- Identify recurring failures
- Measure the impact of software updates
- Evaluate network improvements
- Verify infrastructure upgrades
- Support SLA reporting
- Monitor seasonal traffic behavior

For example, if an application consistently experiences slower response times during peak evening hours, historical reports can help engineers identify capacity constraints and implement corrective actions.

18. Integration with Centra-SD

Centra-Experitest is designed to operate as an extension of Centra-SD, leveraging its centralized management and monitoring capabilities.

While Centra-Experitest focuses on automated application testing, Centra-SD provides the infrastructure required for scheduling, visualization, reporting, and centralized device management.

Integrated Workflow

1. Administrators create automated test scripts.
2. Test schedules are configured in Centra-SD.
3. Digital.ai SeeTest Cloud executes the scripts on real devices.
4. Performance metrics and execution logs are collected.
5. Results are uploaded to Centra-SD.
6. KPI dashboards and visualizations are updated.
7. Alarms are generated if thresholds are exceeded.

8. Engineers analyze reports and investigate issues.
9. Corrective actions are implemented.
10. Subsequent automated tests verify issue resolution.

This integration creates a complete ecosystem for continuous application monitoring and network performance management.

19. Practical Use Cases

Centra-Experitest is suitable for a wide range of operational scenarios across telecommunications, enterprise IT, banking, retail, healthcare, and government sectors.

Mobile Banking Applications

Banks can monitor login functionality, account access, fund transfers, and payment services continuously to ensure uninterrupted customer experience.

E-Commerce Platforms

Retail organizations can verify product search, shopping cart functionality, payment processing, and order confirmation throughout the day.

Video Streaming Services

Media companies can monitor video startup time, playback quality, buffering frequency, and streaming stability across different network conditions.

Government Digital Services

Public service portals can be monitored for application submission, document uploads, payment gateways, and authentication workflows.

Healthcare Applications

Hospitals and healthcare providers can verify appointment booking systems, telemedicine platforms, and patient portals to ensure continuous availability.

Telecom Customer Applications

Telecommunication providers can monitor self-service applications used for bill payment, recharge, service activation, complaint registration, and usage tracking.

Enterprise Business Applications

Organizations can automate testing of HR systems, ERP platforms, CRM solutions, and internal productivity tools to maintain operational continuity.

20. Advantages of Centra-Experitest

Centra-Experitest provides several technical and business advantages that improve service quality while reducing operational complexity.

Continuous Service Monitoring

The platform operates around the clock, ensuring that application performance is continuously evaluated without manual intervention.

Improved Customer Experience

By identifying service degradation before customers encounter issues, organizations can maintain higher levels of user satisfaction and reduce complaint volumes.

Reduced Operational Costs

Automation significantly reduces the need for repetitive manual testing, allowing engineering teams to focus on analysis and optimization.

Real User Simulation

Tests are executed on actual smartphones rather than simulators, providing realistic measurements that accurately reflect end-user experience.

Scalable Deployment

The platform supports execution across thousands of devices, making it suitable for enterprise-scale deployments and nationwide monitoring.

Centralized Visibility

Integration with Centra-SD provides a unified dashboard for monitoring applications, devices, and network performance from a single interface.

Faster Issue Resolution

Detailed logs, automated alerts, and historical analysis help engineering teams identify and resolve issues more quickly.

21. Best Practices

Organizations implementing Centra-Experitest should follow these recommendations to maximize effectiveness.

- Develop reusable automated test scripts.
- Prioritize monitoring of business-critical applications.
- Schedule tests during both peak and off-peak hours.
- Configure meaningful performance thresholds to reduce false alarms.
- Regularly review dashboards and historical reports.
- Maintain updated versions of monitored applications.
- Ensure sufficient device battery and storage for uninterrupted testing.
- Archive execution logs for compliance and audit purposes.
- Periodically validate automated scripts after application updates.

- Integrate findings into continuous improvement and network optimization processes.
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22. Frequently Asked Questions (FAQs)

What is Centra-Experitest?

Centra-Experitest is an automated mobile and web application monitoring platform that integrates with Digital.ai SeeTest Cloud and Centra-SD to continuously evaluate application performance and Quality of Experience (QoE).

What types of applications can be monitored?

The platform supports native mobile applications, hybrid applications, progressive web applications (PWAs), and browser-based web applications.

Does Centra-Experitest use real devices?

Yes. Automated scripts are executed on real smartphones, providing realistic measurements that closely represent actual user experiences.

Can tests be scheduled automatically?

Yes. Administrators can configure recurring schedules that execute automated tests at predefined intervals without manual intervention.

How are failures detected?

The platform compares collected metrics against predefined thresholds. If abnormal behavior is detected, alerts are generated and displayed through Centra-SD.

Can the platform monitor applications continuously?

Yes. Centra-Experitest supports 24×7×365 continuous monitoring of applications and digital services.

23. Glossary

Quality of Experience (QoE): A measure of how users perceive the performance and usability of an application or service.

Automated Test Script: A predefined sequence of actions that simulates user interactions with an application.

Digital.ai SeeTest Cloud: A cloud-based platform that enables automated testing on real mobile devices.

Response Time: The time required for an application or server to respond to a user request.

Application Crash: An unexpected termination of an application due to software or system errors.

Alarm Threshold: A predefined value that triggers an alert when exceeded.

Performance Dashboard: A graphical interface displaying real-time and historical performance metrics.

Regression Testing: Automated testing performed after software updates to verify that existing functionality continues to operate correctly.

24. Conclusion

Centra-Experitest provides a comprehensive solution for continuous monitoring of mobile and web applications by combining automated testing, real-device execution, centralized management, and intelligent performance analysis. Through its integration with Digital.ai SeeTest Cloud and Centra-SD, the platform enables organizations to monitor critical digital services proactively, identify issues before they affect users, and improve overall Quality of Experience.

Its automation capabilities reduce operational costs, while detailed reporting and historical analysis support informed decision-making and continuous service improvement. Whether deployed for telecom services, enterprise applications, banking platforms, or public digital services, Centra-Experitest delivers scalable and reliable monitoring that helps organizations maintain high service availability and customer satisfaction.

25. Relationship Between Centra-Experitest, Centra-SD, and Sigma-ML

Component	Primary Role	Key Function
Sigma-ML	Field Measurement Application	Collects network measurements (signal strength, throughput, VoLTE, protocol logs) on smartphones.
Centra-SD	Centralized Management Platform	Schedules tests, manages devices, stores uploaded logs, visualizes KPIs, and provides historical analysis.
Centra-Experitest	Application Experience Monitoring Platform	Automates testing of mobile and web applications on real devices using Digital.ai SeeTest Cloud, evaluates QoE, and generates proactive alerts.

Integrated Workflow

1. **Sigma-ML** collects wireless network measurements from smartphones.
 2. **Centra-SD** centrally manages devices, schedules tests, and visualizes network performance.
 3. **Centra-Experitest** executes automated application test scenarios on real devices, monitors Quality of Experience, and reports application-level issues.
 4. Together, these products provide a complete solution for **network performance monitoring, application testing, and proactive service assurance**, enabling organizations to optimize both network quality and end-user experience.
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